

Multiple Choice (10 marks)

1. A molecule has a rotational correlation time of 1 ns. At what magnetic field would protons in this molecule have the fastest spin-lattice relaxation? [Use eqns 5.3 and 5.4.]
 - (a) 3.74 T
 - (b) 5.19 T
 - (c) 6.08 T
 - (d) 9.49 T
2. Infrared (IR) spectroscopy is useful for determining certain aspects of the structure of organic molecules because
 - (a) all molecular bonds absorb IR radiation
 - (b) IR peak intensities are related to molecular mass
 - (c) most organic functional groups absorb in a characteristic region of the IR spectrum
 - (d) each element absorbs at a characteristic wavelength
3. Of the following compounds, which has the lowest melting point?
 - (a) HCl
 - (b) AgCl
 - (c) CaCl₂
 - (d) CCl₄
4. Which nuclide has an NMR frequency of 115.5 MHz in a 20.0 T magnetic field?
 - (a) ¹⁷O
 - (b) ¹⁹F
 - (c) ²⁹Si
 - (d) ³¹P
5. Of the following ions, which has the smallest radius?
 - (a) K⁺
 - (b) Ca₂⁺
 - (c) Sc₃⁺
 - (d) Rb⁺

True / False (10 marks)

Answer True or False

6. True or False: The Q-factor for an X-band EPR cavity with a resonator bandwidth of 1.58 MHz is $Q = 6012$.
7. True or False: Tetramethylsilane is used as a reference because its protons are weakly shielded, which affects the chemical shift.
8. True or False: Doubling the volume of a buffer solution significantly increases its pH by diluting the weak acid and conjugate base concentrations.
9. True or False: Silicon is most commonly found in nature as a metallic element, occurring in its pure form in the earth's crust.
10. True or False: CH_3Cl has a ^1H resonance approximately 1.55 kHz downfield from TMS in a 12.0 T magnetic field.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors influencing the equilibrium polarization of an electron and a proton placed in a 1.0 T magnetic field at 300 K, and calculate the ratio of their polarizations.
12. Discuss the process by which the polarization of ^{13}C nuclei can be increased in a 600 MHz spectrometer. How long will it take for the polarization to reach twice its thermal equilibrium value at 298 K, given a T_1 of 5.0 s?

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